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Fundamentals of thery of computation I. – exam 2025.05.26.

0-23: grade 1, 24-31: grade 2, 32-39: grade 3, 40-47: grade 4, 48-60: grade 5

1. True or false? (Put an X in the column of the correct answer.) (2

true	false	
☐	☐	Any homomorphism $h : V_1^* \to V_2^*$ is unambigously determined by giving $h(t) \in V_2^*$ for all $t \in V_1$.
☐	☐	For all formal languages L we have $L^* \setminus \{\varepsilon\} = L^+$.
☐	☐	Let V be an alphabet and $L_1, L_2 \subseteq V^*$ be languages over V. Then $(L_1 \setminus L_2) \cup L$ holds.
☐	☐	All languages that can be recognized by a deterministic finite automaton can b by a left linear grammar.
☐	☐	Regular expressions $ab(\varepsilon + ab)^*$ and $(ab)^*$ describe the same language.
☐	☐	For every nondeterministic finite automaton A there exists a deterministic fini ton A' with the property $L(A') = L(A)$.
☐	☐	$G = (\{S, A\}, \{c, d\}, \{S \to cAd, S \to d, A \to cdA, A \to cc\}, S)$ is a grammar of
☐	☐	Let $E \to b$ a rule in P for the gammar $G = (N, \Sigma, P, S)$, where $E \in N$ and b G is NOT a regular grammar.
☐	☐	If all rules of P for a grammar $G = (N, \Sigma, P, S)$ is one of the schemes $A \to BC$ where $A, B, C \in N, a \in \Sigma$, then G is a grammar in Kuroda normal form.
☐	☐	Intersection of a regular and a context free laguage is context free.
☐	☐	$\{uu^R \mid u \in \{b, c\}^*\}$ is a language of type 3.
☐	☐	All nonterminals of reduced context-free grammars are active and reachable initial symbol.
☐	☐	Every grammar of type 3 is a grammar of type 1 as well.
☐	☐	Let $A = (Q, \Sigma, \delta, q_0, F)$ be a deterministic finite automaton, all of its state are from q_0. Assume, that $q \overset{0}{\sim} r$ and $r \notin F$ hold for some $q, r \in Q$. Then $q \notin F$ ho
☐	☐	The membership problem of context free grammars can be decided in polynom
☐	☐	Let $A = (Z, Q, \Sigma, \delta, z_0, q_0, F)$ be a pushdown automaton. Then $L(A)$ can be ge a grammar in Kuroda normal form.
☐	☐	Let $A = (Z, Q, \Sigma, \delta, z_0, q_0, F)$ be a deterministic finite automaton. If $\vert\delta(z, q, a)\vert$ for some $z \in Z, q \in Q, a \in \Sigma$ then $\delta(z, q, \varepsilon) = \emptyset$.
☐	☐	Let A be a pushdown automaton. Then there exists a nondeterministic finite a A' with the property $L(A') = N(A)$.
☐	☐	Asssume that there is a leaf with label b and its parent with lable A in a deri of a context free grammar $G = (N, \Sigma, P, S)$ then $A \to b \in P$.
☐	☐	Let $L = \{a, ab, bab\}$. Then $L_e = \emptyset$.

NEPTUN: ....

2. Answer the following questions. No explanation is required. (10x2 p

(a) Let $L = \{a^n b^{2k} \mid n \geq 0, k \geq 2\}$.

$L^2 =$ ....

(b) List 3 language operations $\mathcal{L}_2$ not being closed under.

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(c) Let $G_1 = (\{A\}, \{a, b\}, \{A \to bbA \mid ab\}, A)$ and $G_2 = (\{B\}, \{a, b\}, \{B \to aB \mid bl$
be regular grammars. Give a regular grammar generating $L(G_1)L(G_2)$ according t
construction we have learnt. (Don't forget to name the initial symbol.)

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(d) Let $G = (N, \Sigma, P, S)$ be a grammar és $u, v \in (N \cup \Sigma)^*$. Definie what does it mean t
can be derived from u in one step. ($u \Rightarrow_G v$ relation)

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(e) Let $A = (Q, \Sigma, \delta, Q_0, F)$ be a nondeterministic finite automaton. Define $L(A)$, the
guage recognized by A.

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(f) Consider a nondeterministic finite automaton $A = (\{p, q, r\}, \Sigma, \delta, \{p, r\}, \{p, q\})$ (i.e.,
$\{p, q\}$). List all the rejecting states of the deterministic finite automaton A' we have l
to be equivalent with A. (The set of states of A' is the powerset of the set of states o

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(g) Let L be a language of type 3. What is the set of accepting states of the Myhill-Nerod
tomaton A_L^{MN}? (The automaton with the states corresponding to the the suffix langu
wrt. the words over the alphatet of L.)

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(h) Let $A \to BCDXY$ be a rule in a context free grammar $G = (N, \Sigma, P, S)$ with
$\{S, A, B, C, D, X, Y\}$. Give a set of CF rules simulating this rule but having at m
letters on any right hand sides. I.e., apply type 2 length reduction for this rule.

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(i) Give a pushdown automaton rule with the following effect. By an ε-move the autom
puts two letters a on the topmost letter b of the pushdown store and changes the st
the automaton from r to q.

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(j) Place $\mathcal{L}_{\mathrm{detPDA}}$, the class of languages being recognizable by a deteministic pushdow
tomaton in Chomsky's hierarchy of language classes.

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NEPTUN:.........................

(3+3+8+6 points)

(a) Define what is a context free grammar and what is a context free language.

(b) Define type 3 normal form of grammars and state the corresponding normal form theorem.

(c) Introduce how de we eliminate ε rules from a context free grammar to get a grammar equivalent to the original one. (It can be assumed that all terminal letters appear alone on the right hand side of the rules and no rule has a right hand side of length greater than 2.)

(d) Define pushdown automaton (including its components and transition function).